

Sample Questions on Runtime Analysis and Merge Sort Simulation

1

For each of the following functions, determine whether the given statement is correct, and justify your answer using the definition of Big O or Theta.

Q1.1 $f(n) = 3n + 5$. Show that $f(n) \in O(n)$.

Q1.2 $f(n) = 7n^2 + 2n + 8$. Prove that $f(n) \in \Theta(n^2)$.

Q1.3 $f(n) = 4n \log n + 20n + 15$. Show that $f(n) \in \Theta(n \log n)$.

Q1.4 $f(n) = n^3 + 100n^2 + n$. Determine tight upper and lower bounds.

Q1.5 $f(n) = 2^n + 100n^3$. Prove that $f(n) \in \Theta(2^n)$.

Q1.6 $f(n) = 5n + \sqrt{n} + 7$. Prove that $f(n) \in \Theta(n)$.

Q1.7 $f(n) = 10n^2 + 3n \log n + 50$. Determine the smallest $g(n)$ such that $f(n) \in O(g(n))$.

2

Analyze the asymptotic running time of the following code blocks. For each, write the corresponding $f(n)$ (the number of operations as a function of n), and then express it using Big O and Theta notations.

Q2.1 for i = 1 to n:
 print(i)

Find $f(n)$ and classify in Big O and Theta notation.

Q2.2 for i = 1 to n:
 for j = 1 to n:
 print(i, j)

What is the total number of print statements?

Q2.3 for i = 1 to n:
 for j = 1 to i:
 print(i * j)

Determine $f(n)$ and prove it belongs to $\Theta(n^2)$.

Q2.4 `i = 1`
 `while i <= n:`
 `print(i)`
 `i = i * 2`

Find $f(n)$ and show that $f(n) \in \Theta(\log n)$.

Q2.5 `for i = 1 to n:`
 `for j = 1 to n:`
 `for k = 1 to n:`
 `print(i + j + k)`

Compute the asymptotic time complexity.

3

Simulate the merging of two sorted subarrays into one sorted array. Show each comparison and insertion step.

Instructions

- You are given two sorted arrays L and R .
- Simulate the `merge()` process, step-by-step, combining them into one sorted list.
- At each step, show the current elements being compared, and the updated merged list.

Sample Problems

Q3.1 Merge: $L = [1, 4, 6, 8]$, $R = [2, 3, 5, 7]$.

Q3.2 Merge: $L = [2, 5, 9]$, $R = [1, 3, 10]$.

Q3.3 Merge: $L = [1, 6, 7, 8]$, $R = [2, 4, 5, 9]$.

Q3.4 Merge: $L = [10, 20, 30, 40]$, $R = [5, 15, 25, 35]$.

4

Trace the recursive breakdown and merging steps for Merge Sort. Show the recursive divisions, merges, and final sorted array.

Q4.1 Trace Merge Sort on $A = [38, 27, 43, 3, 9, 82, 10]$.

Q4.2 Trace Merge Sort on $A = [5, 2, 4, 6, 1, 3, 7, 8]$.

Q4.3 Trace Merge Sort on $A = [8, 4, 2, 6, 1, 3]$.

Q4.4 Trace Merge Sort on $A = [50, 20, 10, 60, 30, 70]$.